AIGR Focused Hypothesis Report — *eno* (Q88MF9, PSEPK) extracellular region (GO:0005576)

Gene: eno (enolase / phosphopyruvate hydratase), *Pseudomonas putida* KT2440 **UniProt:** Q88MF9 (ENO_PSEPK, reviewed Swiss-Prot), EC 4.2.1.11, 429 aa **Focus:** function-assignment — "eno has extracellular region (GO:0005576)" **Annotation under review:** GO:0005576, IEA, GO_REF:0000044

Executive Judgment

Verdict: Over-annotated (weakly supported for this organism).

The extracellular-region assignment is a purely computational propagation, not an observation about *P. putida* enolase. QuickGO confirms GO:0005576 on Q88MF9 is `IEA / GO_REF:0000044 / ECO:0007322`, mechanically mapped from the UniProtKB **"Secreted"** subcellular-location term. That "Secreted"/"Cell surface" location is itself asserted by **HAMAP rule MF_00318** (ECO:0000255, by sequence similarity) for *every* bacterial enolase matching the rule, carrying the generic note *"Fractions of enolase are present in both the cytoplasm and on the cell surface."* There is **no experimental (non-IEA) evidence** for extracellular or cell-surface enolase in *P. putida*.

The moonlighting surface/secreted enolase phenomenon is real biology, but the primary literature that demonstrates it is confined to **pathogens and parasites** (M. tuberculosis, Mycoplasma, Leptospira, Streptococcus, Bifidobacterium, Giardia), where surface enolase acts as a plasminogen receptor contributing to virulence. *P. putida* KT2440 is a non-pathogenic soil saprophyte; extending a virulence-linked moonlighting location to it by rule is a frequency/paralog-style over-annotation.

The gene product's **direct, primary function is a cytoplasmic Mg^{2+}-dependent glycolytic lyase** (2-phosphoglycerate ⇌ phosphoenolpyruvate). Extracellular localization is at best a context-specific secondary role and, for this organism, unsupported by evidence.

A cross-ortholog audit (Iteration 3) makes the over-annotation explicit: the same `IEA:UniProtKB-SubCell` extracellular term is attached to *E. coli* enolase (a canonical cytoplasmic protein with no surface evidence) but is **absent** from *S. pneumoniae* and *S. aureus* enolases, where surface localization is experimentally established. The annotation is thus **anti-correlated with real biology** and carries no organism-specific information for *P. putida*. No experimental GO:0005576 annotation exists on any enolase examined.

Evidence Matrix

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Co lim
QuickGO record for Q88MF9	database/ computational	Qualifies (provenance)	Is GO:0005576 experimentally supported?	GO:0005576 is IEA, GO_REF:0000044, ECO:0007322, assignedBy UniProt; mapped from "Secreted" SubCell term	*P. putida* KT2440	Hi co is r de ex ba
HAMAP MF_00318 (UniRule)	computational/ evolutionary	Qualifies	Where does "Secreted/Cell surface" come from?	Rule blanket-asserts Cytoplasm + Secreted + Cell surface + generic note for all matching bacterial enolases (ECO:0000255)	Pan-bacterial rule	Hi ex pr no or sp
P 27569900 (Mtb enolase)	direct assay / localization	Competing / context	Is surface enolase demonstrated?	Enolase is surface-exposed, high-affinity plasminogen binder; C-terminal Lys required	*M. tuberculosis* (pathogen)	Str pa no tra to
P 35337383 (Mycoplasma hyorhinis)	direct assay / localization	Competing / context	Surface enolase & host binding	Flow cytometry confirms surface enolase; binds plasminogen/ fibronectin; mediates cytoadhesion	Pathogen	Str sp *Ps*
				Enolase secreted by unknown	Pathogen	De lea

Citation	Evidence type	Supports/ Refutes/ Qualifies	Claim tested	Key finding	Context	Co lim
P 27989763 (Leptospira)	direct assay / localization	Competing / context	Secretion + surface reassociation	mechanism, reassociates with membrane, binds plasminogen/ complement regulators		sec pa on
P 24319673 (Streptococcus review)	review	Competing / context	Moonlighting surface plasminogen binders	"Surface displayed cytoplasmic proteins with enzymatic activities (moonlighting proteins)" as a class	Streptococcus	Re vir fra
P 23872606 (moonlighting review)	review	Qualifies	Why do glycolytic enzymes appear on surface?	Surface localization tied to virulence/ benefit; secretion/ localization non-classical and regulated	Bacteria (general)	Re sup "se co sp rea
UniProt Q88MF9 keywords/ comments	database	Supports core function	What is the primary function?	Cytoplasm (HAMAP), Glycolysis, Lyase, Mg-binding; FUNCTION = 2-PG⇌PEP, essential for glycolysis	*P. putida*	Hi cyt fur

GO Curation Implications

Lead (requires curator verification): do NOT accept GO:0005576 as a supported/core annotation for eno in this review; treat it as non-core / candidate for removal or downgrade.

- The annotation is a mechanical `IEA / GO_REF:0000044` SubCell mapping seeded by a pan-bacterial HAMAP rule, not evidence that *P. putida* enolase directly resides in the extracellular region. Under AIGR guidance, a rule-propagated CC term with no organism-specific support and a plausible over-annotation mechanism should not be marked as a demonstrated function.
- **Cellular component that IS supported:** GO:0005737 (cytoplasm) — consistent with the enzyme's characterized function and the HAMAP cytoplasm assertion.
- The related IEA companions GO:0009986 (cell surface) and the "Secreted" keyword ride on the same rule and inherit the same weakness; if the curator removes/downgrades GO:0005576, the cell-surface term should be handled identically.
- **Molecular function / Biological process (the true core):** phosphopyruvate hydratase activity (GO:0000287 Mg binding; GO:0004634 phosphopyruvate hydratase activity) and glycolytic process (GO:0006096) — these are the informative, defensible annotations.

Recommended action label: **NOT_SUPPORTED / mark as over-annotation** for GO:0005576 (pending curator confirmation), with rationale "IEA rule propagation of a pathogen-associated moonlighting location to a non-pathogen; no organism-specific evidence."

Mechanistic Scope

- **Direct molecular activity being annotated by CC term:** physical presence of the enolase polypeptide in the extracellular region.
- **Established direct activity of the gene product:** cytoplasmic, homodimeric, Mg^{2+}-dependent lyase catalyzing 2-PG ⇌ PEP in glycolysis/gluconeogenesis.
- **The extracellular claim is downstream/secondary at best:** in organisms where it is real, surface enolase is a *moonlighting* plasminogen receptor reached by a non-classical (signal-peptide-independent) export route and re-association with the envelope — a virulence-linked secondary role, not the enzyme's primary catalytic job.

- For *P. putida* there is no demonstrated export, no demonstrated surface display, and no functional consequence reported; the CC term reflects rule inference only.

Conflicts and Alternatives

- **Rule/database carry-over (most likely explanation):** HAMAP MF_00318 applies Secreted/Cell surface uniformly, so the annotation reflects the rule, not the strain.
- **Organism mismatch:** all supporting experimental data are from pathogens/parasites that exploit host plasminogen; *P. putida* is an environmental non-pathogen with no such host interface, weakening biological plausibility.
- **Mechanistic requirement not evidenced:** surface plasminogen binding depends on a C-terminal lysine and specific internal Lys sites (e.g.,

  P 27569900

  P 28770921);

  whether Q88MF9 has the relevant C-terminal/internal Lys determinants was not tested here and would still not establish localization in *P. putida*.
- **No competing experimental annotation** places enolase extracellularly in *Pseudomonas*; the only conflict is between an automatic prediction and the absence of evidence.

Knowledge Gaps

1. **Organism-specific localization.** Checked: QuickGO/UniProt evidence codes — all IEA. Gap: no *P. putida* surface-proteome / secretome / OMV proteomics located enolase. Resolution: strain-specific surface biotinylation, secretome LC-MS/MS, or immuno-EM/flow cytometry.
2. **Export mechanism.** Enolase is leaderless; whether *P. putida* has any route to externalize it is unknown. Resolution: fractionation + membrane-integrity controls to exclude lysis artifacts.
3. **Sequence determinants.** Not analyzed here: presence/absence of the C-terminal and internal lysine plasminogen-binding motifs in Q88MF9. Resolution: alignment to experimentally validated surface enolases; but positive motif ≠ localization proof.

Discriminating Tests

- **Surface-shaving / biotinylation proteomics** of intact *P. putida* KT2440 cells with strict cytoplasmic-contamination controls (e.g., absence of other abundant cytoplasmic markers) — the single most decisive experiment.
- **Secretome and OMV LC-MS/MS** under relevant growth conditions to see if enolase is externalized at all.
- **Flow cytometry / immuno-EM** with anti-enolase antibody on non-permeabilized cells.
- **Comparative bioinformatics:** align Q88MF9 to validated surface enolases (Mtb, Strep) to check C-terminal Lys / plasminogen-binding motif conservation — supportive context, not proof.

Curation Leads (require curator verification)

- **Action change:** downgrade/remove GO:0005576 (extracellular region) as a supported annotation for eno; classify as **over-annotation from IEA rule propagation**. Apply the same treatment to GO:0009986 (cell surface) and the "Secreted" keyword.
- **Retain / promote:** GO:0005737 (cytoplasm, CC), GO:0004634 (phosphopyruvate hydratase activity, MF), GO:0000287 (magnesium ion binding, MF), GO:0006096 (glycolytic process, BP).
- **Provenance to verify in the record:**
- QuickGO: `GO:0005576 | IEA | GO_REF:0000044 | ECO:0007322 | assignedBy UniProt`.
- HAMAP MF_00318 subcellular-location note: *"Fractions of enolase are present in both the cytoplasm and on the cell surface"* (rule-level, ECO:0000255) — the origin of the term.
- **Candidate reference snippets (pathogen context, to justify "not transferable"):**
- P 27569900 : *"Enolase, a glycolytic enzyme, has long been studied as an anchorless protein present on the surface of many pathogenic bacteria that aids in tissue remodeling and invasion by binding to host plasminogen."*

- P 23872606 :
  "They localise to the bacterial surface to take on additional activities, which have been hypothesised to contribute to bacterial virulence or bacterial benefit."
- **Suggested curator question:** Is there any *P. putida* (or close-relative) surface/secretome proteomic dataset placing enolase extracellularly? If not, the term stands only on rule inference.
- **Suggested experiment:** surface biotinylation + LC-MS/MS on intact KT2440 cells with cytoplasmic-leakage controls.

Cross-Ortholog Evidence-Code Audit (Iteration 3, computed)

Question: is the extracellular/cell-surface GO term on bacterial enolases ever experimental, and does its presence track real surface biology?

Enolase ortholog	GO:0005576 / GO:0009986 present?	Evidence	Experimentally surface-localized?
Q88MF9 — *P. putida* (query)	Yes	IEA:UniProtKB-SubCell	No data
P0A6P9 — *E. coli*	**Yes**	IEA:UniProtKB-SubCell	**No** (canonical cytoplasmic / RNA-degradosome)
P64075 — *M. tuberculosis*	Yes	IEA:UniProtKB-SubCell	(moonlighting reported)
P77972 — *Bifidobacterium*	Yes	IEA:UniProtKB-SubCell	Yes (P 24840471)
P9WNV9 — *M. tuberculosis*	**No CC term**	—	**Yes** (P 27569900)
Q8DR60 — *S. pneumoniae*	**No CC term**	—	**Yes** (well-established PgR)
P0A4G2 — *S. aureus*	**No CC term**	—	Yes (reported)

Reading: the extracellular/cell-surface term is applied by rule to enolases where surface display is **not** experimentally supported (*E. coli*), yet is **absent** from several enolases where it **is** experimentally proven (*S. pneumoniae*, *S. aureus*, one Mtb entry). The annotation is therefore **anti-correlated with real biology** — classic frequency-bias/rule over-annotation that carries no organism-specific information for *P. putida*. Every occurrence found is `IEA:UniProtKB-SubCell`; **no experimental GO:0005576 annotation exists on any enolase examined.**

Sequence-Feature Provenance (Iteration 2, computed)

Direct check of Q88MF9's physical plausibility for extracellular localization (public FASTA):

Feature	Q88MF9 (*P. putida*)	E. coli enolase P0A6P9 (cytoplasmic)	Interpretation
Length	429 aa	432 aa	Canonical bacterial enolase
N-terminus	`MAKIVDIKGREVLDSRGNPTVEA…`	`MSKIVKIIGREIIDSRGNPTVEA…`	Conserved cytoplasmic enolase start; **no signal peptide**
N-term hydrophobicity (aa 1–25)	0.40, no cleavable h-region	similar	No Sec/Tat secretion signal
C-terminal residue	**Gly** (`…RGRAEFRG`)	Ala	**No C-terminal Lys** (a determinant for some surface enolases' plasminogen binding, e.g., Mtb P 27569900)
Lys content	6.8% (29 K)	8.6%	Unremarkable

Reading: the sequence is that of a classical cytoplasmic glycolytic enzyme with no export signal and without the C-terminal lysine motif linked to plasminogen-binding surface enolases. Any extracellular presence would require a non-classical export route that has not been demonstrated in *P. putida*. This is supportive computational context, reported conservatively — absence of a motif is not proof of non-localization, but it provides **no positive support** for the GO:0005576 hypothesis and is consistent with the over-annotation conclusion.

Limitations

This assessment relied on public UniProt/QuickGO/HAMAP records and PubMed; local `*-bioinformatics` analyses were intentionally withheld. Targeted PubMed queries for *P. putida* enolase surface/secretome returned no hits, which is itself informative (absence of organism-specific evidence) but not absolute proof of non-localization. No sequence-motif analysis of Q88MF9's lysine determinants was performed.
